RADAR-ASSISTED BORDER SURVEILLANCE FOR UAV DETECTION

A PROJECT REPORT

Submitted by

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ABSTRACT

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Low altitude UAV intrusion is an emerging security threat in remote border areas, whereby constant human manpower and aerial surveillance are costly in terms of operations and are influenced by weather conditions. In this paper, a low-budget radar-emulation prototype sector-scanning surveillance structure based on an ESP32 microcontroller and an ultrasonic time-of-flight (TOF) sensor is described. The system conducts an indexed 360 ° environmental scanning in eight angular sectors with the help of a stepper-motor-driven sensing assembly. Local processing of distance measurements is done by use of temporal filtering to remove spurious reflections and minimise false alarms. Authenticated cases of intrusion are sent over Wi-Fi to a remote monitoring station where a visualisation program written in MATLAB is used to give a real-time polar plot and a log of events. Under conditions of low-altitude intrusion experiments, it has been experimentally validated that the directional detection has high reliability at a radius of 4 m, an average alert latency of 0.82 s, and 95% detection consistency in repeated tests. The intended architecture is a scalable prototype platform of radar-assisted border surveillance studies, focusing on modularity, power-efficiency, and cost-efficient implementation on resource-constrained deployments.

Keywords: Border surveillance, UAV intrusion detection, ESP32, ultrasonic sensing, directional scanning, IoT monitoring, MATLAB visualization.

TABLE OF CONTENTS

TABLE OF CONTENTS

CHAPTER No.	TITLE	PAGE No.
	ABSTRACT	**i**
	LIST OF TABLES	**vi**
	LIST OF FIGURES	**vii**
	LIST OF ABBREVIATION	**ix**
1	**INTRODUCTION**	**1**
	1.1 BACKGROUND	2
	1.2 NEED FOR THE PROJECT	2
	1.3 PROBLEM IDENTIFICATION	3
	1.4 OBJECTIVE	3
	1.5 ORGANIZATION OF PROJECT	3
	1.6 SUMMARY	4
2	**LITERATURE SURVEY**	**5**
	2.1 INTRODUCTION	5
	2.2 LITERATURE	6
	2.2.1 BIRD VS DRONE CLASSIFICATION USING MOTION FEATURES AND RANDOM FOREST	6
	2.2.2 DOPPLER-BASED HUMAN–VEHICLE CLASSIFICATION USING FMCW RADAR	6
	2.2.3 IOT SMART FENCE USING ESP32 AND BLYNK	7
	2.2.4 CNN-BASED UAV DETECTION USING SENSOR FUSION	7
	2.2.5 ULTRASONIC SCANNER CONTROLLED BY MATLAB AND MICROCONTROLLER	8

	2.2.6 IOT-BASED OBJECT DETECTION RADAR WITH MATLAB VISUALIZATION	8
	2.2.7 SURVEY ON ANTI-DRONE SYSTEMS AND CHALLENGES	9
	2.2.8 AI-BASED BORDER TRESPASSER CLASSIFICATION	9
	2.3 SUMMARY	9
3	**PROPOSED SYSTEM**	**10**
	3.1 INTRODUCTION	10
	3.2 SYSTEM OVERVIEW	10
	3.3 EMBEDDED CONTROL ALGORITHM	11
	3.4 WORKING PRINCIPLE OF SURVEILLANCE SYSTEM	13
	3.5 HARDWARE ARCHITECTURE	15
	3.6 SOFTWARE ARCHITECTURE AND DATA PROCESSING	16
	3.7 DEPLOYMENT CONSIDERATIONS AND SCALABILITY	17
	3.8 SUMMARY	17
4	**HARDWARE DESCRIPTION**	**18**
	4.1 INTRODUCTION	18
	4.2 HARDWARE ARCHITECTURE	18
	4.3 POWER SUPPLY UNIT	19
	4.4 ULTRASONIC SENSING MODULE	20
	4.5 ESP32 EMBEDDED CONTROL UNIT	21
	4.6 STEPPER MOTOR AND MOTOR DRIVER INTERFACE	22
	4.7 DISPLAY AND USER INTERFACE MODULE	24

	4.8 PHYSICAL IMPLEMENTAION AND MOUNTING	24
	4.9 SUMMARY	25
5	**SOFTWARE DESCRIPTION**	**26**
	5.1 INTRODUCTION	26
	5.2 SOFTWARE ARCHITECTURE	26
	5.3 EMBEDDED SOFTWARE IMPLEMENTATION USING ARDUINO IDE	28
	5.4 SERIAL COMMUNICATION AND DATE TRANSMISSION	29
	5.5 MATLAB-BASED SIGNAL PROCESSING AND RADAR VISUALIZATION	29
	5.6 INTRUSION DETECTION USING RANDOM FOREST ALGORITHM	30
	5.7 CLOUD INTEGRATION USING BLYNK PLATFORM	31
	5.8 SUMMARY	31
6	**RESULT AND DISCUSSION**	**32**
	6.1 INTRODUCTION	32
	6.2 EXPERIMENTAL SETUP AND TESTING CONDITION	32
	6.3 DIRECTIONAL DETECTION ANALYSIS	33
	6.4 CLOUD IOT MONITIORING THROUGH BLINK PLATFORM	38
	6.5 PERFORMANCE EVALUATION OF THE PROPOSED SYSTEN	39
	6.6 COMPARATIVE ANALYSIS WITH EXISTING SURVEILLANCE SYSTEMS	39
	6.7 DISCUSSION	41
	6.8 SUMMARY	41
7	**CONCLUSION AND FUTURE WORK**	**42**

7.1 CONCLUSION 42

7.2 FUTURE WORK 43

REFERENCE 45

PUBLICATIONS 49

LIST OF TABLES

LIST OF TABLES

TABLE No.	TITLE	PAGE No.
6.1	PERFORMANCE ANALYSIS OF PROPOSED SYSTEM	39
6.2	COMPARISION BETWEEN EXISTING AND PROPOSED SYSTEM	40

LIST OF FIGURES

LIST OF FIGURES

FIGURE No.	TITLE	PAGE No.
3.1	BLOCK DIAGRAM OF THE PROPOSED SYSTEM	11
3.2	FLOWCHART OF THE PROPOSED SYSTEM	12
4.1	PROTEUS CIRCUIT DIAGRAM OF THE COMPLETE HARDWARE SYSTEM	19
4.2	ULTRASONIC SENSOR MODULE (HC – SR04 TYPE)	21
4.3	ESP32 DEVELOPMENT BOARD	21
4.4	STEPPER MOTOR	23
4.5	MOTOR DRIVER	23
4.6	16x2 LCD INTERFACE	24
4.7	HARDWARE PROTOTYPE	25
5.1	SOFTWARE EXECUTION FLOW	27
5.2	ARDUINO IDE IMPLEMENTATION OF SECTOR SCANNING	28
5.3	MATLAB RADAR VISUALIZATION INTERFACTE	30
6.1	DETECTION IN NORTH AND SOUTH DIRECTIONS (MATLAB and LCD)	34

6.2	DETECTION IN EAST AND WEST DIRECTIONS (MATLAB and LCD)	35
6.3	DETECTION IN NORTH-WEST AND SOUTH-WEST DIRECTIONS (MATLAB and LCD)	36
6.4	DETECTION IN NORTH-EAST AND SOUTH EAST DIRECTIONS(MATLAB and LCD)	37
6.5	CLOUD IoT BLINK PLATFORM	38

LIST OF ABBREVIATIONS

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ACRONYMS	ABBREVIATIONS
CNN	**C**onvolutional **N**eural **N**etwork
FMCW	**F**requency **M**odulated **C**ontinuous **W**ave
GPIO	**G**eneral **P**urpose **I**nput/**O**utput
GPU	**G**raphics **P**rocessing **U**nit
IDE	**I**ntegrated **D**evelopment **E**nvironment
LCD	**L**iquid **C**rystal **D**isplay
MATLAB	**Mat**rix **Lab**oratory
PWM	**P**ulse **W**idth **M**odulation
RF	**R**adio **F**requency
SPI	**S**erial **P**eripheral **I**nterface
ToF	**T**ime **o**f **F**light
UART	**U**niversal **A**synchronous **R**eceiver **T**ransmitter
UAV	**U**nmanned **A**erial **V**ehicle
USB	**U**niversal **S**erial **B**us
Wi-Fi	**Wi**reless **Fi**delity
GUI	**G**raphical **U**ser **I**nterface
IC	**I**ntegrated **C**ircuit
LED	**L**ight **E**mitting **D**iode

CHAPTER 1
INTRODUCTION

CHAPTER 1
INTRODUCTION

Unmanned aerial vehicles (UAVs) or drone are gaining popularity in the army and among civilians. Illegal operation of UAVs in sensitive or restricted zones has become a significant security concern. The detection and classification of such threats have been extensively studied with the help of advanced radar-based detection and classification methods, including machine learning-based aerial target classification and Doppler-based detection methods [13], [5].

However, the traditional radar detection systems are effective, albeit very expensive and complicated to use in short and low-cost applications [24], [8]. Conversely, recent advancements in deep learning and sensor fusion have enhanced the accuracy of UAV detection, but they demand a lot of computation power and sophisticated hardware [29], [30]. To address these drawbacks, the proposed study will concentrate on a cheap radar-emulation system based on ultrasonic sensors.

The system should be able to scan the various areas, detect objects in real time, and give information on distance and position. Other related systems based on ultrasonic sensors and microcontroller-based platforms have been found to be successful in object detection and MATLAB based visualization [9], [19]. Moreover, microcontrollers like ESP32 are promising in IoT-based surveillance systems and have been demonstrated to perform well in remote monitoring and security systems [4].

The suggested system will combine these technologies to offer a low-cost and efficient platform of detecting UAVs at short ranges and patrolling the boundaries.

1.1 BACKGROUND

The fast development of the Unmanned Aerial Vehicle (UAV) technology has made drones extremely popular in the civilian and military fields. Some of the most common applications of UAVs include surveillance, aerial photography, monitoring agricultural activities, disaster management, and delivery of parcels. Despite the numerous benefits associated with these applications, the abuse of drones has also brought up serious concerns in the areas of privacy, safety, and national security. The UAVs might be a source of threat to unauthorized intrusion into controlled areas like airports, military bases, and governmental buildings. Thus, there is increasing demand of good and effective systems to identify such intrusions over short distances.

The common traditional radar systems are applied to detect aerial objects and operate based on the electromagnetic waves and the advanced signal processing methods. Although these systems offer precise long-range detection, they are costly, sophisticated, large and demand substantial power and infrastructure unsupportive. These restrictions render them inappropriate in small scale or short distance monitoring. Ultrasonic sensors, however, offer an easy and inexpensive approach to short-range detection of objects with sound waves of high frequency. A radar-emulation system can be simplified by combining ultrasonic sensing with servo-based sector scanning and used to detect and display the position and distance of objects in real time.

1.2 NEED FOR THE PROJECT

The rising cases of unauthorized UAV incursions into restricted zones raise significant security and safety issues. The traditional radar systems are expensive and not applicable when monitoring small areas or small distances. There is also the need to have a compact and economical detection system that has the ability to cover limited areas.

A simplified radar-emulation model is useful in learning the principles of radar in a more practical and less expensive way. Thus, this project has to engineer a robust short-range intrusion detection system on the basis of the ultrasonic technology.

1.3 PROBLEM IDENTIFICATION

The key issues that were highlighted in this project are:

- The rise of UAV intrusion in restricted and sensitive zones.
- Expensive and difficult to use traditional radar systems in short range uses.
- Absence of small-scale intrusion detection systems that are compact and economical.
- Inability to effectively locate and monitor small UAVs in small-range systems.

1.4 OBJECTIVE

This project aims at designing and developing an ultrasonic radar-emulation system in the sector to detect intrusion of short-range UAVs. It tries to identify the objects with the help of ultrasonic sensing and display the real-time distance and angular position. The system will be low-priced, small-sized, and applicable to small-scale security applications.

1.5 ORGANIZATION OF PROJECT

The structure of the project consists of the introduction, literature survey, system design, hardware and software implementation, results, and conclusion. The chapters elaborate on how the proposed system was developed and how it works.

CHAPTER 1: This chapter is concerned with the introduction and the overview to have bare idea on the project.

CHAPTER 2: This chapter is concerned with the Literature survey in order to have a better understanding of relevancy towards the improvement of the proposed work.

CHAPTER 3: This chapter explains the project of the planned work to be done and the technology applied in order to better the project.

CHAPTER 4: In Chapter 4, the hardware requirements and their description and test results are explained to understand more.

CHAPTER 5: This chapter addresses the description of the software that will be utilized in this technology.

CHAPTER 6: This chapter is concerned with the project outcome and discussion.

CHAPTER 7: This chapter describes the conclusion obtained and scope of future work on the project.

1.6 SUMMARY

The chapter presented the idea of the Sector-Scanning Ultrasonic Radar-Emulation System of short-range UAV intrusion detection. It has discussed the history of the research, defined the problem, outlined the purpose, and the necessity of creation of a low cost and effective detection system.

CHAPTER 2
LITERATURE SURVEY

CHAPTER 2
LITERATURE SURVEY

2.1 INTRODUCTION

The last chapter presented the idea of UAV detection and border surveillance systems and how the problem of unauthorized UAVs in sensitive areas has been on the rise due to the security threats they pose. There was also the need to have automated surveillance systems to help the security personnel in identifying aerial intrusion.

Conventional border surveillance systems are based on manual surveillance, costly radar surveillance, or large-scale sensor networks, which is inappropriate in low cost and small-scale applications [15], [25]. As wireless sensor networks (WSNs), IoT, and machine learning have improved, new surveillance solutions have become more efficient and scalable [6], [22].

This chapter provides a review of the literature on the UAV detection, radar surveillance systems, machine learning-classification, and IoT-based monitoring solutions. Some scholars have particularly concentrated on radar signal processing methods like micro-Doppler analysis and range-Doppler mapping in order to enhance the accuracy of small UAV detection [8], [7].

Random Forest, CNNs, and SVMs are widely utilized machine learning and deep learning models that are used to classify targets based on radar [13], [29], [12], [11]. Also, AI-surveillance has been used to identify and categorize border trespassers and increase real-time decision-making [27], [14]. The IoT-based systems incorporated sensors, microcontrollers and communication protocols have made it possible to implement smart surveillance and remote monitoring applications [4], [26].

Recent research also addresses multi-sensor fusion and edge computing methods to enhance the accuracy of the detection and decrease the latency of a surveillance system [6], [28]. Moreover, anti-drone systems and UAV detectors have been widely researched and multiple design issues and implementation aspects have been outlined [1], [17]. In spite of these developments, most of the existing systems are expensive, complicated or computationally involved.Thus, there is a research gap in the creation of a low-cost, efficient, and real-time radar tracking system of a UAV that can be used in small-scale operations. This vacuum is what encourages the suggested Radar-Assisted Border Surveillance System where an ESP32 microcontroller, ultrasonic sensing, and MATLAB-based real-time monitoring are used to detect objects.

2.2 LITERATURE

2.2.1 Bird vs Drone Classification Using Motion Features and Random Forest

This study aims at identifying the difference between drones and birds based on motion traits derived out of the surveillance radar images. The authors applied machine learning methods, specifically, a Random Forest classifier, to characterize radar echoes patterns and movement characteristics. The system works with radar signals to determine distinctive flight patterns of drones and birds to enhance the accuracy of classifying in surveillance. The system however needs high-resolution radar data and intensive computing capability. It also requires advanced radar equipment which makes the systems more complex and expensive to implement in small scale applications.[1].

2.2.2 Doppler-Based Human–Vehicle Classification Using FMCW Radar

The article presents a classification system that employs Doppler spectrum properties acquired through the Frequency Modulated Continuous Wave (FMCW) radar sensors to differentiate between humans and cars.

The analysis of Doppler signatures emitted by moving objects with the use of machine learning algorithms allows the accurate classification of these signatures in the surveillance environment. The study demonstrates the usefulness of radar-based sensing with machine learning in the object recognition task. The method shows enhanced reliability of detection even in the changing environmental conditions. The system however, needs special FMCW radar equipment and intricate signal processing code. These demands render such implementation less applicable to low-cost embedded surveillance systems.[2].

2.2.3 IoT Smart Fence Using ESP32 and Blynk

This paper is a smart fence system based on the IoT that is used in remote monitoring and security. The suggested system also involves the use of NodeMCU ESP32 and sensors to monitor the unauthorized access and provide the alerts in the form of the Blynk IoT. The system allows real time monitoring via mobile devices and gives remote control.

The study shows the possibility of the IoT technology in the form of cost-effective security systems in the residential and industrial regions. Though the system is remote accessible and low power consuming, it is mainly concentrated on ground level intrusion detection. It does not have sophisticated sensing of aerial surveillance and fails to tackle the issue of the UAV detection in border conditions. [3].

2.2.4 CNN-Based UAV Detection Using Sensor Fusion

This research paper suggests a UAV detection model based on Convolutional Neural Networks (CNN) sensor fusion method. Various sensor signals including radar and optical data are combined to enhance precision of the detection and minimize the false alarms. The deep learning algorithm is conditioned to identify UAV targets with an integrated sensor characteristic, which has strong detection capabilities.

The experimental findings indicate that sensor fusion method can greatly improve the classification reliability in a complex setting. Nevertheless, deep learning models have a high training data and computational needs. Also, the design of a system that combines a number of sensors adds complexity and cost to the system, which is hard to achieve with a small embedded system. [4].

2.2.5 IoT-Based Object Detection Radar with MATLAB Visualization

This study explains how an ultrasonic scanning system was developed, which was operated using the MATLAB and a microcontroller platform. The system works based on an ultrasonic sensor that is mounted on a rotating system that scans the surrounding environment and identifies obstacles. The resulting data on distance is sent to the MATLAB to be visualized and analyzed. The experiment shows that low-cost sensors and microcontrollers may be utilized in order to simulate radar-like scanning capability. The visualization interface allows viewing the results of objects detected in real time. Nonetheless, the system primarily deals with indoor obstacle detection and short-range scanning. It is not about UAV detection or border surveillance application in particular.[5].

2.2.6 Survey on Anti-Drone Systems and Challenges

This paper introduces a radar-inspired object detector, which combines the use of microcontrollers in sensing and MATLAB in visualization. The system captures the environment through sensors attached on a rotating platform, which are then used to send the data to MATLAB to be graphically represented. The design will show how embedded systems and software visualization tools can be used together to develop real time monitoring interfaces. The system has a radar-like display that is graphical in nature to detect objects and the distance. The implementation is however based on simple object detection as opposed to aerial target identification. It is also restricted in its range of detection and its accuracy when applied in real-life border surveillance applications. [6].

2.2.7 AI-Based Border Trespasser Classification

The paper is a detailed overview of the current anti-drone technologies and detection systems. It talks about various detection systems such as radar, acoustic sensors, radio frequency monitoring and vision systems. The paper also reviews other countermeasure methods applied in the drone mitigation.

The survey identifies the merits and flaws of various sensing technologies and how they can be deployed. It states that there is the necessity of combined and economical detection systems which can be used in various settings. The majority of the systems that have been examined in the research are however intricate infrastructures as well as costly equipment and are hence not accessible to small-scale surveillance purposes.[7].

2.2.8 AI-Based Border Trespasser Classification

This study examines artificial intelligence methods of categorizing illegal crossers at the border using sensor data and surveillance inputs. Machine learning algorithms are used to study the trends and identify unauthorized movements in border regions. The purpose of the system is to support security forces by offering automated detection and classification functions.

The study proves the viability of AI-based surveillance in enhancing border security performance. But the suggested solution is mostly concentrated on human trespasser and on-ground threats. It is not specifically concerned with the issues of UAV intrusion detection in aerial surveillance case.[8].

2.3 SUMMARY

The studies show that radar, machine learning, and IoT are widely used for surveillance and object detection, improving accuracy but often being costly and complex. The proposed system overcomes these issues by using a low-cost ESP32-based radar prototype with ultrasonic sensing and MATLAB visualization for 360° UAV monitoring.

CHAPTER 3
PROPOSED SYSTEM

CHAPTER 3
PROPOSED SYSTEM

3.1 INTRODUCTION

The former chapter has already assessed the current technologies of detecting UAVs, radar surveillance and short-range sensing methods, which have weaknesses that include high cost of deployment, limited coverage and subject to environmental perturbation. These observations resulted in the need to have a small, energy saving and scalable border monitoring solution.

This chapter outlines the design and construction of the planned system, which was based on a pole-mounted sector-scanning surveillance system with an ESP32 microcontroller, an ultrasonic time-of-flight (TOF) sensor as a radar prototype, and a rotating mechanism based on a stepper motor as an indexed 360deg scan.

3.2 SYSTEM OVERVIEW

The proposed system is designed as an aerial and surface intrusion radar-supported prototype system of border surveillance within a short-range perimeter. It comprises of three key subsystems, including sensing and scanning unit, embedded processing and communication unit, and remote monitoring and visualization unit.

The sensing unit is fitted with an ultrasonic ranging module on a stepper motor shaft, which allows angular movement to be done under control in 45deg steps. This has a 360deg field divided into eight indexed sectors, North, North-East, East, South-East, South, South-West, West, North-West. The overall architecture of the proposed radar-assisted surveillance system is shown in Fig. 3.1.

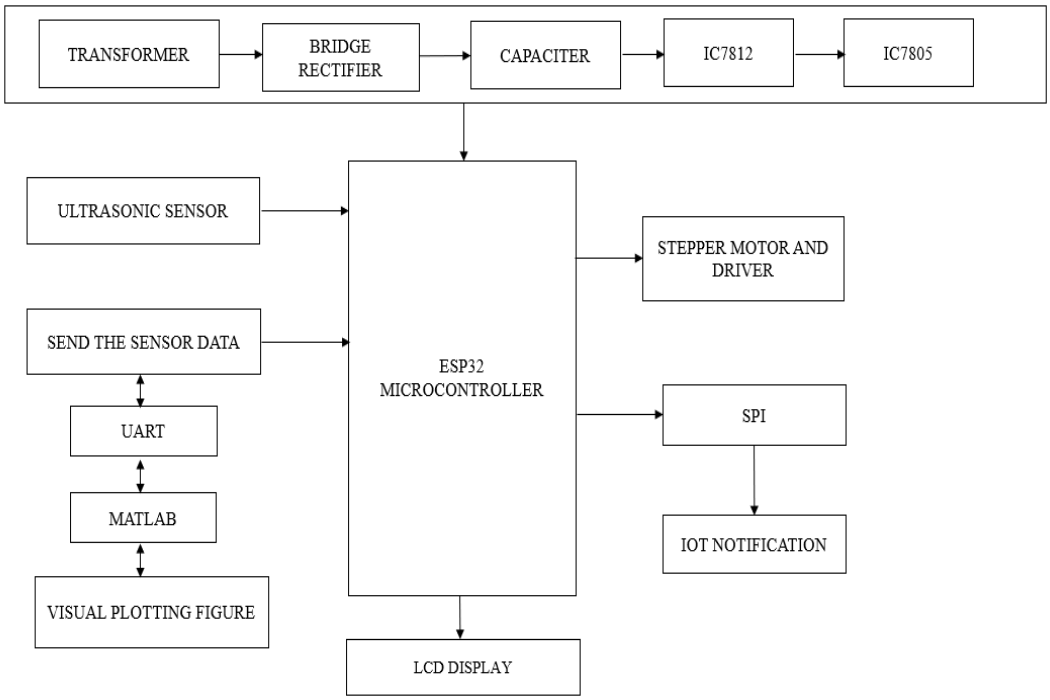

Fig. 3.1. Block Diagram of the Proposed System

The embedded processing unit focuses on the ESP32 microcontroller that will be used to control the rotation of the motors, initiate ultrasonic pulses, measure the time of echo return, filter the local and process wireless data transmission. The remote monitoring unit will consist of a MATLAB interface that will be at the receiving end of real-time data, visualization of sector-wise detection status as well as recording of time-stamped intrusion events.

Here should be presented a system block diagram that shows how the ESP32, stepper motor driver, ultrasonic sensor, wireless module, and monitoring station are connected to each other.

3.3 EMBEDDED CONTROL ALGORITHM

The radar-assisted surveillance prototype suggested is executed with the help of an organized embedded control algorithm deployed on the ESP32 microcontroller.

The algorithm organizes the work of sensing, decision-making, communication, and mechanical rotation to provide constant monitoring of the surrounding environment in sectors. The flowchart representing the working process of the proposed system is shown in Fig. 3.2.

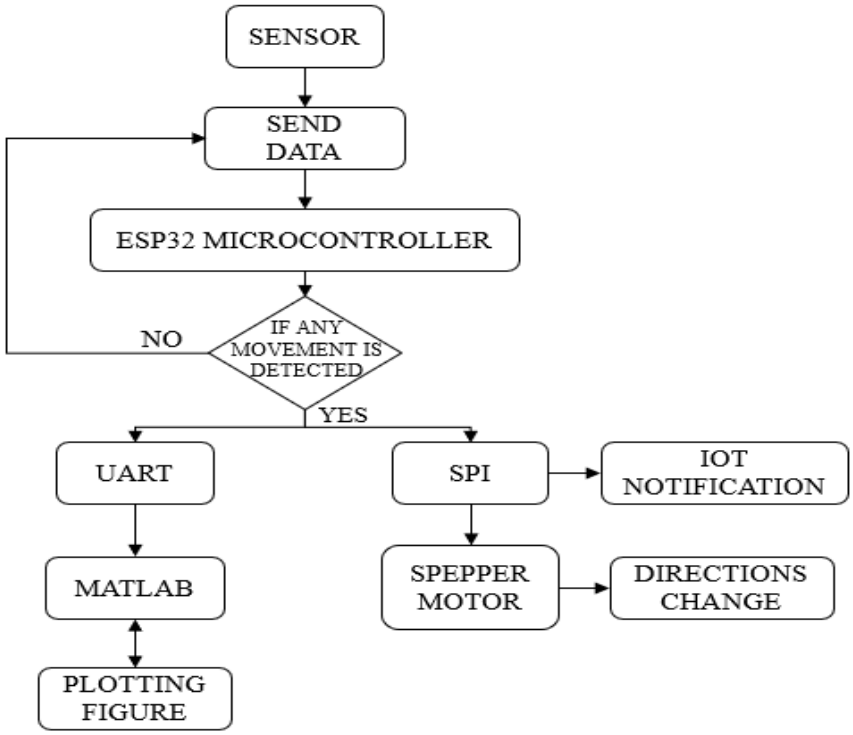

Fig. 3.2. Flowchart of the Proposed System

Algorithm: Intrusion Detection and Sector Scanner algorithm

Step 1: Boot the system and initialize ESP32 microcontroller.

Step 2: Program ultrasonic sensor, stepper motor driver, UART communication interface, SPI interface, IoT module and LCD display.

Step 3: Connect to remote monitoring station (MATLAB interface) and make the IoT notifications to be connected wirelessly.

Step 4: Turn on the ultrasonic sensor in order to get distance information of the sector.

Step 5: Send the sensed data to ESP32 microcontroller to be processed.

Step 6: Process the sensor data obtained to define whether some movement or intrusion of objects is detected.

Step 7: In case the movement is not detected, keep scanning by holding the monitoring loop.

Step 8: If movement is detected:

- Send the data of the detectors to the MATLAB interface through UART to visualize them in real-time.
- Send alert data to the internet of things notification module through SPI communications.
- Indicate the status detected on the LCD.

Step 9: Turn the stepper motor in order to shift the scanning direction to the next predefined sector.

Step 10: Repeat the sensing and detection of all the eight sectors.

Step 11: Repeat the scanning cycle to offer sustained surveillance of the borders.

3.4 WORKING PRINCIPLE OF THE SURVEILLANCE SYSTEM

The system suggested is designed under the principle of active sensing of time-of-flight (ToF). An ultrasonic sensor in this prototype has the capabilities of a short-range radar. The sensor emits a pulse of acoustic frequency and then measures the time that the echo of the object takes to bounce back.

Where t is the echo returned time measured and v is the velocity of sound in air (about 343 m/s at room temperature) then the distance d to the object is calculated as:

$$d = \frac{v \cdot t}{2} \qquad \text{----------(3.1)}$$

The propagation of the signal of the signal in both directions is divided by two. To be practical in the implementation in microcontroller-based systems, the equation is commonly written in microseconds:

$$d = \frac{v \cdot t_{\mu s}}{2 \times 10^6} \qquad \text{----------(3.2)}$$

Where $t_{\mu s}$ the width of the pulse to be echoed in micro seconds.

The ESP32 rotates the sensor and moves it to the angular positions in a sequence. Several distance samples are taken at every sector to minimize measurement uncertainty.

A validation system that uses threshold is used to check whether an object is within the stipulated intrusion radius (4 m in the prototype). And the distance measured meets the following:

$$d \leq d_{threshold} \qquad \text{----------(3.3)}$$

An intrusion is marked as possible. The temporal filtering is also used to reduce the false detections due to environmental noise, vegetation movement, or small animals.

Consecutive measurements are only detected and sent to the monitoring station when they are over a confidence threshold.

3.5 HARDWARE ARCHITECTURE

The proposed system hardware system is a compact pole-mounted unit that incorporates sensing, actuation, processing, and communication units.

The active ranging module is the ultrasonic sensor. Though the traditional radar uses electromagnetic waves in its operation, the ultrasonic sensor is used in this prototype to simulate radar-based detection in a way that is economical.

The sensor has the trigger and echo pins connected with the ESP32 GPIO ports in order to generate pulses and measure the time.

A motor driver module controls the stepper motor by accepting the control signals of ESP32. The motor has an indexed rotation of 45deg steps and provides fine angular location of each sector.

The angular position θ_n for sector n is defined as:

$$\theta_n = 45^\circ \times n, n = 0,1,2,\dots,7 \qquad \text{----------(3.4)}$$

This discrete angular indexing enables structured scanning and consistent directional mapping.

The ESP32 microcontroller acts as the main controller and it does signal acquisition, computation, motor control and Wi-Fi communication.

Its in-built wireless feature also removes the necessity of adding more communication modules to the system, which simplifies the system and, in turn, makes it consume less power. There should be a detailed circuit diagram of the ESP32 connection with ultrasonic sensor, motor driver, power supply and LCD in this section. Also, a picture of the hardware prototype can be provided to demonstrate physical implementation.

3.6 SOFTWARE ARCHITECTURE AND DATA PROCESSING

The embedded firmware running on the ESP32 and a MATLAB-based interface to monitor the ground station is used to implement the software architecture.

The firmware has a cyclic scanning algorithm. Once the system has been initialised, the ESP32 spins the sensor to a set sector, initiates ultra sonic ranging, calculates the distance, verifies the measurement and records the reading.

This is incrementally repeated with all the eight sectors creating a single scan cycle.

The scan time T_{scan}can be approximated as:

$$T_{scan} = 8 \times (T_{rotate} + T_{measure} + T_{process}) \quad \text{----------(3.5)}$$

where $T_{rotate}$is time of indexing the motor, $T_{measure}$ is the time of data acquisition in the ultrasonic echo, and $T_{process}$ is the time of data validation and communication. The experimental findings have shown that the average alert latency is about 0.82 s.

The ESP32 sends the sector index, measured distance, and timestamp of an intrusion event via Wi-Fi through IoT communication protocols on confirmation of an intrusion event. In the monitoring station, the incoming data are fed into MATLAB which then plots real-time polar or cartesian plot with directional markers. The event logs are kept to be analyzed.

Control commands to the ESP32 can also be sent in MATLAB to recalibrate the ESP32, scan a particular sector, or adjust the threshold. This section should contain a software flow chart of the embedded algorithm and MATLAB communication protocol.

3.7 DEPLOYMENT CONSIDERATIONS AND SCALABILITY

The suggested architecture will operate in remote border areas that need constant surveillance. The pole mounted system improves line of sight and minimizes ground level interference.

Sequential scanning as opposed to continuous full-angle sweeping produces energy efficiency. The modular architecture enables many units to be implemented in longer border ranges to create a distributed surveillance system. Each node will be able to act separately and relay data to a central monitoring station.

In spite of the current implementation using ultrasonic sensing in short range application (up to 4 m), the platform can be modified to use microwave radar modules on long range applications. Signal processing and sector-indexing approach can still be used and so the prototype is a scalable platform towards the next-generation radar-assisted UAV detection systems.

3.8 SUMMARY

In this chapter, the entire design and implementation of the suggested radar-assisted border surveillance prototype was introduced. The system combines ESP32 microcontroller, sector-indexed stepper motor mechanism and an ultrasonic time-of-flight sensor to result in structured 360deg monitoring. Distance estimation mathematical modelling, threshold-based intrusion validation, hardware interfacing, and MATLAB based visualization were explained. The suggested architecture offers a low cost, energy efficient, and scalable platform to short-range UAV intrusion detection. The second chapter reveals the experimental configuration, performance analysis, and analysis of the results of the prototype implemented under control test conditions.

CHAPTER 4
HARDWARE DESCRIPTION

CHAPTER 4
HARDWARE DESCRIPTION

4.1 INTRODUCTION

The former chapter gave the general system architecture and methodology of the proposed Radar-Assisted Border Surveillance System in the UAV Detection that explains the sector-scanning concept, signal acquisition mechanism and control scheme used in detecting short range aerial targets. Continuing in the idea and algorithmic design, this chapter is a full description of the hardware architecture that brings into practice the proposed system.

This chapter is meant to describe the design, component choice, electrical design and physical implementation of the surveillance prototype. It describes the power conditioning phase, sensing subsystem, embedded controller architecture, motor actuation mechanism, and display interface. Theoretical background, working principles and mathematical formulations applicable in ultrasonic ranging, and in the positioning of stepper motors are also discussed. This hardware description is the basis of the experimental validation and performance analysis which is described in the next chapter.

4.2 Hardware Architecture

The hardware system proposed is that of a pole mounted sector scanning detection unit that can monitor the environment 360deg. The architecture comprises of five significant subsystems:

- Regulated Power Supply Unit
- Ultrasonic Radar-Emulation Sensing Module
- ESP32-Based Embedded Control Unit
- Stepper Motor and Driver Interface
- LCD Display and Alert Module

The relationship between these subsystems creates a closed loop scanning and detection system. The ultrasonic sensor is attached to a stepper motor that allows angular movement in 45deg steps to move through eight preset sectors (North, North-East, East, South-East, South, South-West, West, North-West). The ESP32 microcontroller coordinates the rotation of the motor and the distance, processes the echo signal and renews the display interface. The experimental setup of the proposed system is shown in Fig. 4.1.

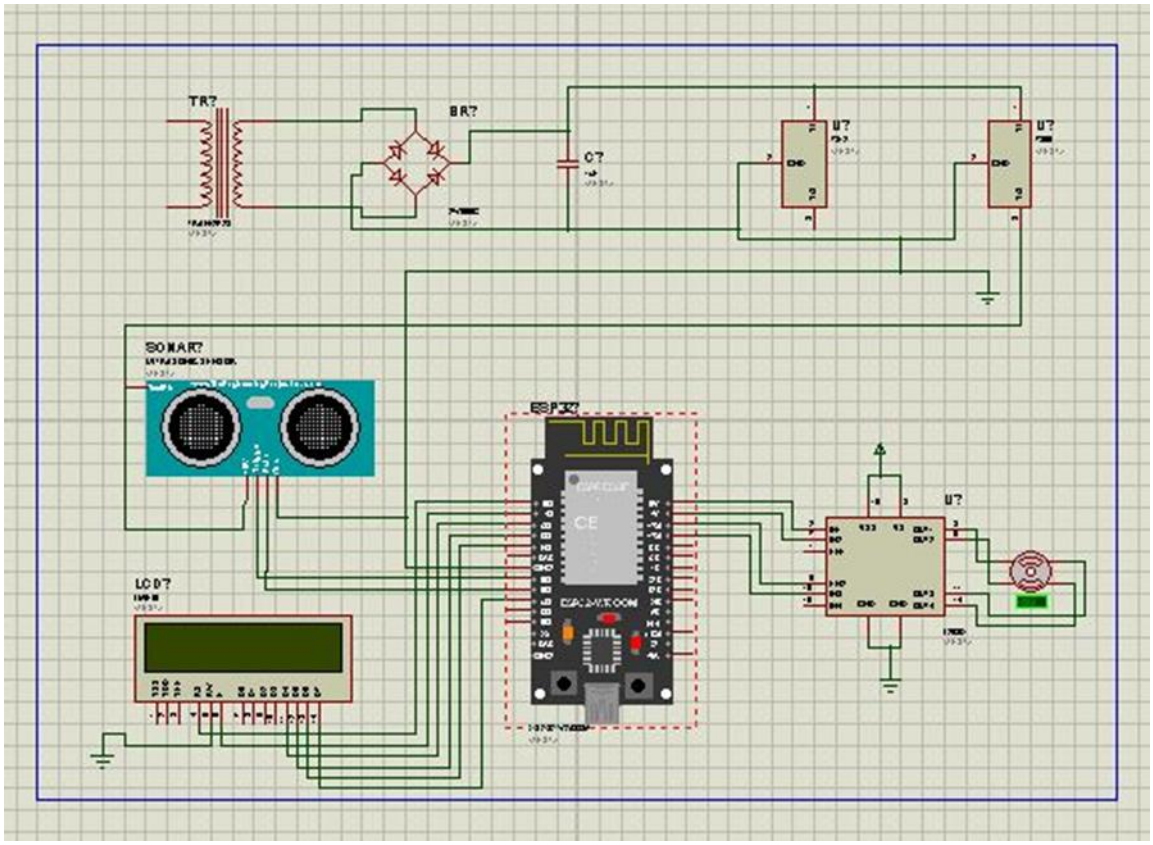

Fig. 4.1. Proteus Circuit Diagram of the Complete Hardware System

4.3 Power Supply Unit

The embedded surveillance system requires a stable and noise free power supply in order to operate reliably. The power supply component comprises a step-down transformer, bridge rectifier, smoothing capacitor and voltage regulation stage.

Transformer is used to convert the 230 V AC mains supply to a lower value AC supply that can be rectified. The full-wave bridge rectifier is a device that changes the AC waveform into a pulsating DC. The rectified output voltage VDCafter full-wave rectification is given as:

$$V_{DC} = \frac{2V_m}{\pi} \qquad \text{----------(4.1)}$$

In which V_m is the maximum secondary transformer voltage.

To minimize ripple voltage a filter capacitor is used across the rectifier output. Ripple factor rf full-wave rectifier with capacitive filtering is estimated as:

$$r = \frac{1}{\sqrt{3} f R_L C} \qquad \text{----------(4.2)}$$

Where, f= supply frequency, R_L= load resistance, C= filter capacitance.

A linear voltage regulator is used to maintain a constant voltage of 5 V to power logic circuit boards and 12 V to power motors. Switching noise produced by the motor driver is not allowed to pass through the decoupling capacitors which are properly decoupled.

4.4 Ultrasonic Sensing Module

The system uses an ultrasonic transceiver module which functions at a frequency of about 40 kHz in order to simulate short-range radar functionality. The sensor is in the form of a pair of transmitters and receivers. When triggered, the transmitter sends a high-frequency acoustic pulse that is reflected by an obstacle (as a UAV) and reflected by air and comes back to the receiver as an echo. The time-of-flight (ToF) principle is the distance measurement principle. Assuming that trep is the round-trip time of the ultrasonic pulse and v is the speed of sound in air (at 25 o C, the speed of sound is about 343 m/s), the distance dto the target would be given by:

$$d = \frac{v \times t}{2} \qquad \text{----------(4.3)}$$

The forward and return propagation are divided by two. The ESP32 calculates the distance and the pulse width of the echo in terms of the internal timers and uses the result to calculate the distance. The sensor can give good measurements between a range of 3 cm to 3.5 m with a standard angular beam width of about 15deg.

In order to achieve angular scanning, the ultrasonic module is firmly fixed on the shaft of the stepper motor. This mechanical integration has provided sector-wise surveillance that resembles radar sweeping. The ultrasonic sensor module (HC-SR04 type) used for distance measurement is shown in Fig. 4.2.

Fig. 4.2. Ultrasonic Sensor Module (HC-SR04 Type)

4.5 ESP32 Embedded Control Unit

The ESP32 microcontroller is the 32-bit dual-core CPU of the system, which has built-in Wi-Fi and Bluetooth support. It conducts coordinated actions of the sensing and actuation modules and the processing and display output of data. The ESP32 development board used in the proposed system is shown in Fig. 4.3.

The ESP32 has a clock frequency of up to 240 MHz and has several GPIO pins, hardware timers, PWM modules and ADC channels. In the proposed system, it will carry out the following operations:

- Produces trigger pulses of the ultrasonic sensor.
- Measures echo pulse duration by use of internal timers.
- Calculates the object distance by using ToF equations.
- Stepping motor- rotates through driver signals.
- Displays sector-wise values of the distance on LCD.

The firmware is built by a sequential scanning algorithm. Each angular position measures the distance data and then the sensor records these values and the motor moves to the next sector. The mean end-to-end update time is dependent on the pulse duration, the delay of a motor step and the processing time.

Fig. 4.3. ESP32 Development Board

4.6 Stepper Motor and Motor Driver Interface

The scanning system is implemented with the help of a stepper motor that is driven by a specific motor driver IC. A stepper motor is used to convert digital pulse sequences into discrete angular motion. When the motor has a step angle ths, the number of steps Nrequired to take one complete revolution is:

$$N = \frac{360^\circ}{\theta_s} \qquad \text{----------(4.4)}$$

The motor in the proposed design will be programmed to turn in steps of 45deg sector divisions. A motor driver is an electronic circuit or device to control the working of the electric motors as an interface between the microcontroller and the motor. The motor driver boosts the control signals that are of low power to the motor and since motors need more current and voltage than a microcontroller can provide, the motor driver drives the motor effectively. It facilitates activities like turning the motor on, off and changing its direction and speed. H-bridge configurations are common with common motor drivers, including the L293D and L298N, which can be operated in two directions. These motors are commonly applied in such applications as robotics, automation systems, and devices using IoT applications where the accuracy of motor control is crucial. The stepper motor used for sector scanning mechanism is shown in Fig. 4.4.

Sector accuracy is determined by the angular resolution and repeatable positioning is achieved by controlled stepping without the need to have feedback sensors (open-loop control). Mechanical rigidity is ensured to avoid the misalignment of rotation. The motor driver used to control the stepper motor is shown in Fig. 4.5.

Fig. 4.4. Stepper Motor

Fig. 4.5. Motor Driver

4.7 Display and User Interface Module

It uses a 16x2 alphanumeric LCD module which is used to display real-time measurements of distance and sector. The LCD uses a 4-bit parallel mode of communication in order to minimize the use of GPIO. The ESP32 transmits data in the form of ASCII coded messages to show sector direction and distance value. The display improves situational awareness by displaying real-time object detection in each direction scanned. Also, alert conditions can be programmed as threshold-based conditions and they can be set to run warning indicators when the detected distance is lower than predefined limit. The 16×2 LCD interfacing used for displaying output is shown in Fig. 4.6.

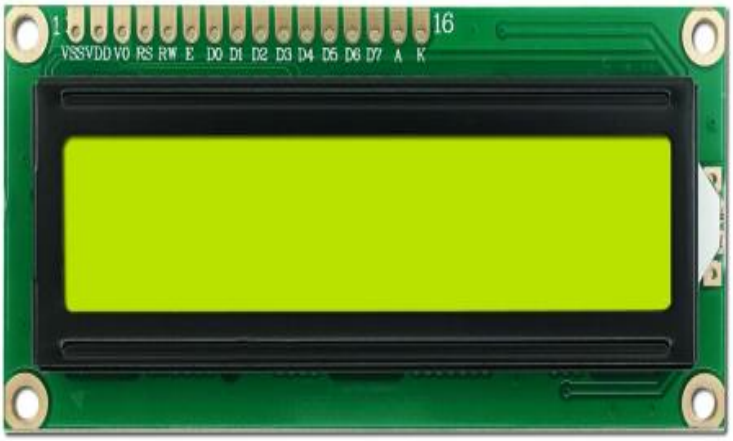

Fig. 4.6. 16×2 LCD Interface

4.8 Physical Implementation and Mounting

The hardware parts are glued on a dot board and are mounted on a rigid support structure. The ultrasonic sensor is attached firmly on the rotating shaft of the stepper motor so as to maintain a constant position. The entire system is contained in a protective casing that is fit to operate in border surveillance environments. Particular consideration is made on the grounding techniques and decoupling strategies to reduce the measurement noise.The hardware prototype of the proposed system is shown in Fig. 4.7

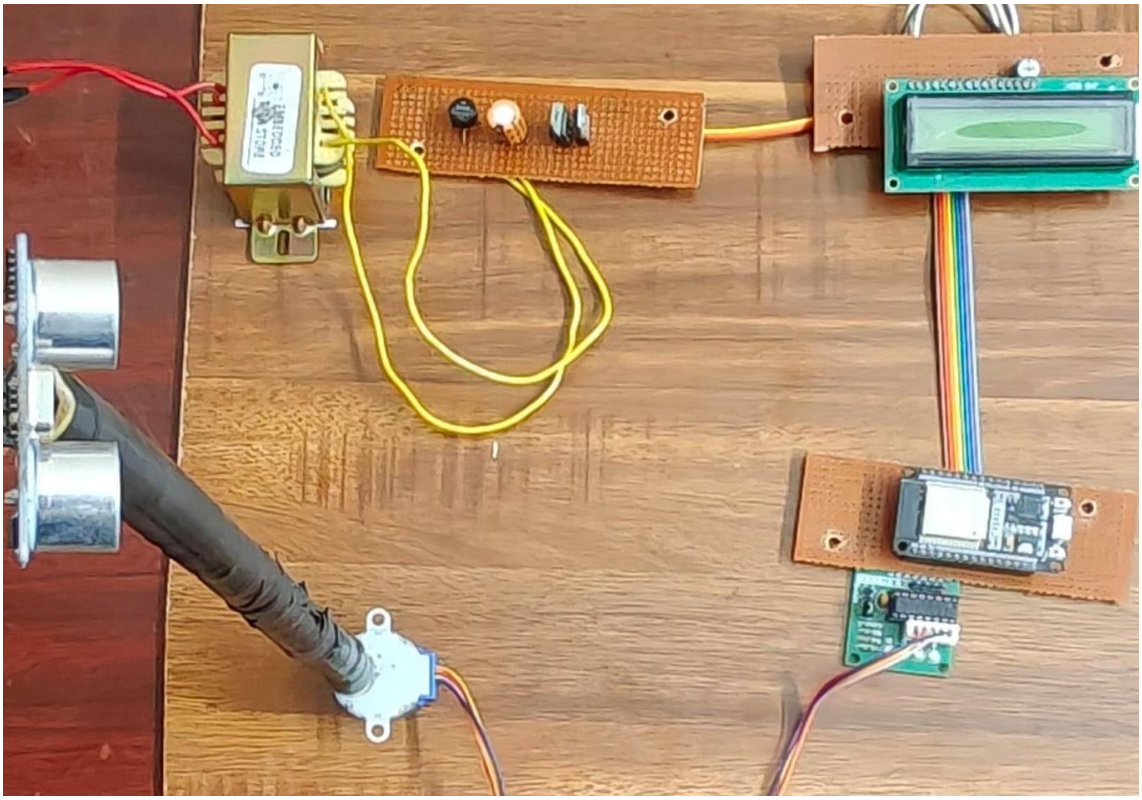

Fig. 4.7. Hardware Prototype

4.9 SUMMARY

In this chapter, the specifics of the hardware design and implementation of the Radar-Assisted Border Surveillance System to detect UAVs were presented. It talked about the controlled unit of power supply, ultrasonic time-of-flight sensing mechanism, embedded control architecture based on ESP32, stepper motor scanning subsystem, and display interface. It contained mathematical formulations of rectification, ripple reduction, distance calculation, and angular stepping to give it theoretical foundation. The given hardware platform defines the hardware implementation of the suggested radar-emulation system and forms the framework on which the experimental analysis and the performance evaluation will be addressed in the following chapter.

CHAPTER 5

SOFTWARE DESCRIPTION

CHAPTER 5
SOFTWARE DESCRIPTION

5.1 INTRODUCTION

The last chapter has introduced the hardware architecture and system integration of the proposed Radar-Assisted Border Surveillance System in UAV Detection and the ultrasonic sensing unit, sector scanning system using servos, microcontroller interface, and wireless communication unit were discussed. Although the hardware subsystem is the physical basis of the surveillance platform, the software architecture of the system determines the operational intelligence and real-time decision-making ability of the system.

The chapter outlines the entire software implementation of the proposed system, which involves embedded programming in Arduino IDE, signal acquisition and processing in MATLAB, intrusion classification in the Random Forest algorithm, and remote monitoring in the Blynk Cloud system. The logical flow of the execution, communication protocols, sector-scanning mechanism, signal validation, and the process of producing alerts are explained in the chapter. The discussion is in a systematic and technical way to demonstrate clearly how the software components interact to bring about credible intrusion detection of UAVs.

5.2 Software Architecture

The new surveillance system is proposed to be built on a distributed software architecture that will have three major layers:

- Embedded Control Layer (Arduino IDE)
- Signal Processing and Visualization Layer (MATLAB)
- Cloud Monitoring and Alert Layer (Blynk Cloud)

The real-time sensor triggering, sector rotation control, and preliminary transmission of data are under the control of the embedded controller. The process of acquiring data in MATLAB is based on the serial communication method, data processing and verification of the signals received, the implementation of the classification model of the random forest, and the creation of radar-like visualization. Blynk Cloud platform is capable of remote alert notification and live monitoring on Wi-Fi. The software execution flow of the proposed system is shown in Fig. 5.1.

This software works in a cyclic scanning model where the area under survey is broken down into angular sectors. Every sector is scanned, analysed and validated in that order before proceeding to the next sector. This replicates a standard radar sweep System.

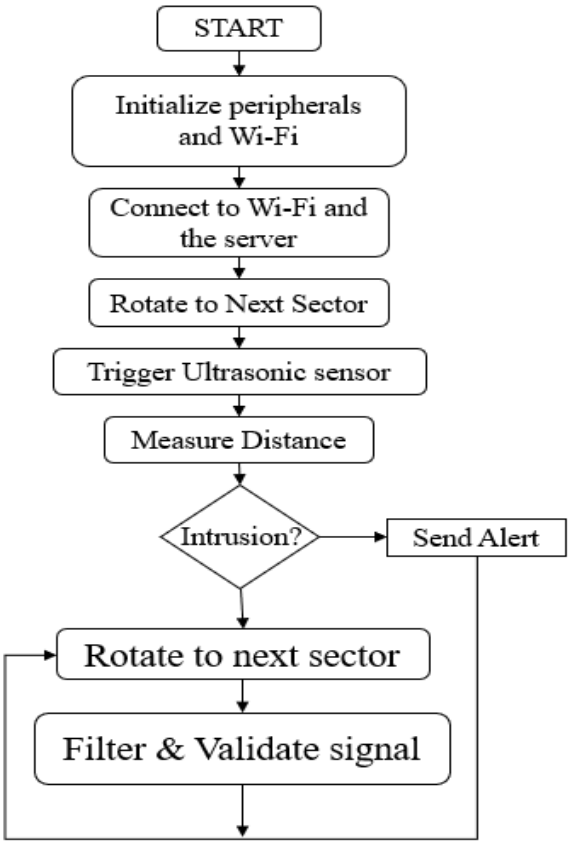

Fig. 5.1. Software Execution Flow

5.3 Embedded Software Implementation Using Arduino IDE

The integrated software is built on the Arduino IDE platform. The program is designed in the form of startup code and endless loops. Once the system is started, the peripheral devices such as ultrasonic sensors, servo motor drivers and Wi-Fi module are started. The microcontroller initiates communication with the Blynk Cloud server with the help of predefined authentication credentials. After the connectivity, the scanning is started. The sector-scanning logic is implemented using controlled servo rotation. The servo motor is programmed to rotate incrementally across predefined angular positions covering the surveillance range. At each angular position, the ultrasonic sensor is triggered to emit acoustic pulses. The echo response is captured and converted into distance measurements.

When the distance that is measured is lower than a predetermined security range, the microcontroller triggers an intrusion event. This data, as well as the data on angles and distance, is sent to MATLAB in serial communication and to the Blynk Cloud in order to be notified remotely. This process is repeated continuously in a loop by the firmware, and thus, provides 360-degree surveillance coverage that is not interrupted. The Arduino IDE implementation of the sector scanning mechanism is shown in Fig. 5.2.

Fig. 5.2. Arduino IDE Implementation of Sector Scanning

5.4 Serial Communication and Data Transmission

Serial communication is used to exchange reliable data between the microcontroller and MATLAB. Angular position and the distance measurement as a formatted data string are sent through the Arduino transmissions. MATLAB is constantly monitoring the communication port that is configured with a specified baud rate.

Data structure ensures that the data transmitted is interpreted in a synchronized manner at the receiver side. A data packet contains the information on sector angle and the distance of the object measured.

This data stream is received by MATLAB which then stores the data to be used in future. Low latency and real-time responsiveness of the communication protocol is also necessary in dynamic intrusion detection applications.

5.5 MATLAB-Based Signal Processing and Radar Visualization

The main signal processing and visualization environment is MATLAB. Once serial communication has been established, MATLAB creates a radar display interface graphic.

The interface is simulated to appear like a polar radar display, with angular position used to indicate servo position and radial distance is used to indicate the detected object range.

When sensor data is fed into MATLAB, signal filtering and validation is done to remove spurious reflection and noise artifacts. Filtering stage enhances stability in detecting the transient or false echoes such that unwarranted alerts are not activated.

The radar plotting is transformed into Cartesian coordinates to plot the validated data. The graphical interface is dynamically changing in real time and showing the detected objects with their respective direction and distance. The MATLAB-based radar visualization interface is shown in Fig. 5.3.

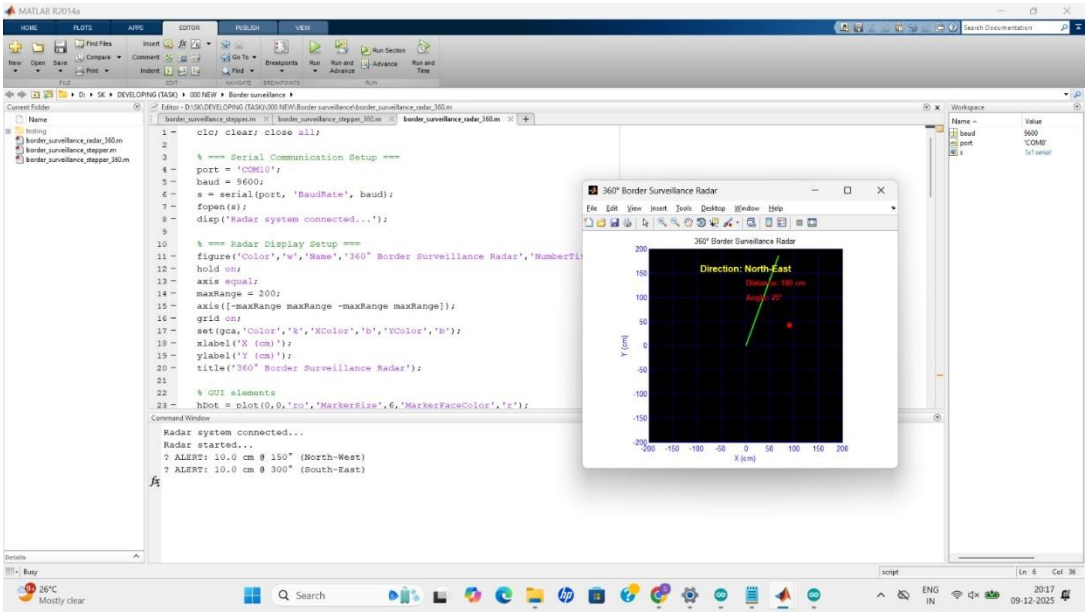

Fig. 5.3. MATLAB Radar Visualization Interface

5.6 Intrusion Detection Using Random Forest Algorithm

In a bid to increase the reliability of detection, the system incorporates a machine learning-based classification step with the help of the Random Forest algorithm in MATLAB. The training of the Random Forest model is performed with the help of labeled data of distances and sectors of normal background and intrusion conditions.

The algorithm works on the principle of building several decision trees and combining each one of them to generate a final classification decision. As every incoming validated data sample is analyzed by the trained model, it is identified whether the sample belongs to a valid environmental reflection or a possible UAV intrusion.

Random Forest is an ensemble-based method, which enhances noise and over-fitting resistance. This also contributes greatly to the accuracy of intrusion detection over and above the basic threshold-based approaches. An intrusion event is confirmed by the classifier, and this triggers an alert generation mechanism in the system.

5.7 Cloud Integration Using Blynk Platform

The Blynk Cloud platform is connected to offer remote monitoring. After the microcontroller notices or is notified by the MATLAB processing of intrusion, a signal of notification is transmitted over Wi-Fi to the Blynk server.

The cloud dashboard shows the status of intrusion, sector information and real-time distance values. Push notification may be created to inform security staff immediately. This also makes it possible to gain remote situational awareness without having to be physically close to the surveillance system.

Cloud integration will change the system into a single radar emulator to an IoT-compatible smart surveillance system that can be applied in border monitoring.

5.8 Summary

The entire software design and implementation of the Radar-Assisted Border Surveillance System to UAV Detection were presented in this chapter. The inbuilt firmware coded in Arduino IDE does the sector-based scanning and sensor control and MATLAB does the data acquisition, signal filtering, radar visualization and intrusion classification using random forest. The Blynk Cloud can be used to monitor and be notified of an alert. These aspects are combined to create a smart, unified surveillance system. The chapter will then give the experimental findings and performance of the implemented system in different intrusion situations.

CHAPTER 6
RESULT AND DISCUSSION

CHAPTER 6
RESULT AND DISCUSSION

6.1 Introduction

In this chapter, the experimental findings and analytical report on the proposed Radar-Assisted Border Surveillance System of UAV Detection are provided. The main aim of this chapter is to test the operational performance of the developed prototype under the control and real-time testing conditions. System performance is measured through directional detection capability, response latency, consistency in detecting, integration with clouds and comparative performance with the current surveillance methods.

The last chapter outlined the software description and implementation structure of the system, embedded programming on the Arduino IDE platform to scan the sector and control the sensor, the data transmission protocols of the serial communication, signal processing and radar view in MATLAB, intrusion classification through the deployment of the Random Forest algorithm, and integration of the IoT based on the Blynk Cloud platform. Based on that software architecture, this chapter confirms the synchronous operation of the embedded, processing, and cloud subsystems, and evaluates the usefulness of the combined surveillance software in identifying low-altitude UAV intrusions in short-range border watch operations.

6.2 Experimental Setup and Testing Conditions

The proposed system was experimentally justified in a semi-open field test which simulated a short-range border surveillance scenario. The ultrasonic sensor was attached to a stepper motor to scan through fixed angular stages. Though the prototype is capable of supporting eight sectors in a 360deg range, the current experimental validation was done in six major directions to examine directional reliability and visualization consistency.

The ESP32 microcontroller was coded to conduct cyclic scanning and scan and get distance measurements at each sector. The embedded firmware was applied to implement temporal filtering to reduce fluctuations caused by noise and eliminate the spurious reflections. When the conditions of intrusion were confirmed in the range of the threshold radius of 4 meters, the system produced three outputs concurrently:

- Local LCD directional display
- Remote MATLAB polar visualization
- Cloud event update via Blink IoT platform

Monotonic intrusion tests were performed at various angular areas to test the consistency of detection, latency of response, and stability of the system.

6.3 Directional Detection Analysis

The system was able to identify intrusions in all the eight directions: North, North-East, East, South-East, South, South-West, West, and North-West. Ultrasonic time-of-flight was used to measure distance and intrusion was verified with respect to a set threshold. The detections were shown in the MATLAB interface in the form of directional plots where a line corresponded to the position of the object in a certain sector. Fig. 6.1 shows a clear visualization and comparison of detections in North and South directions with LCD outputs. Similarly, Fig. 6.2 represents the correct detection of East and West directions. Fig. 6.3 illustrates the detection of North-East and South-East directions, while Fig. 6.4 shows the detection results for North-West and South-West directions. The system response time was approximately 200 ms and display was steady at a time when continuous scanning was going on.

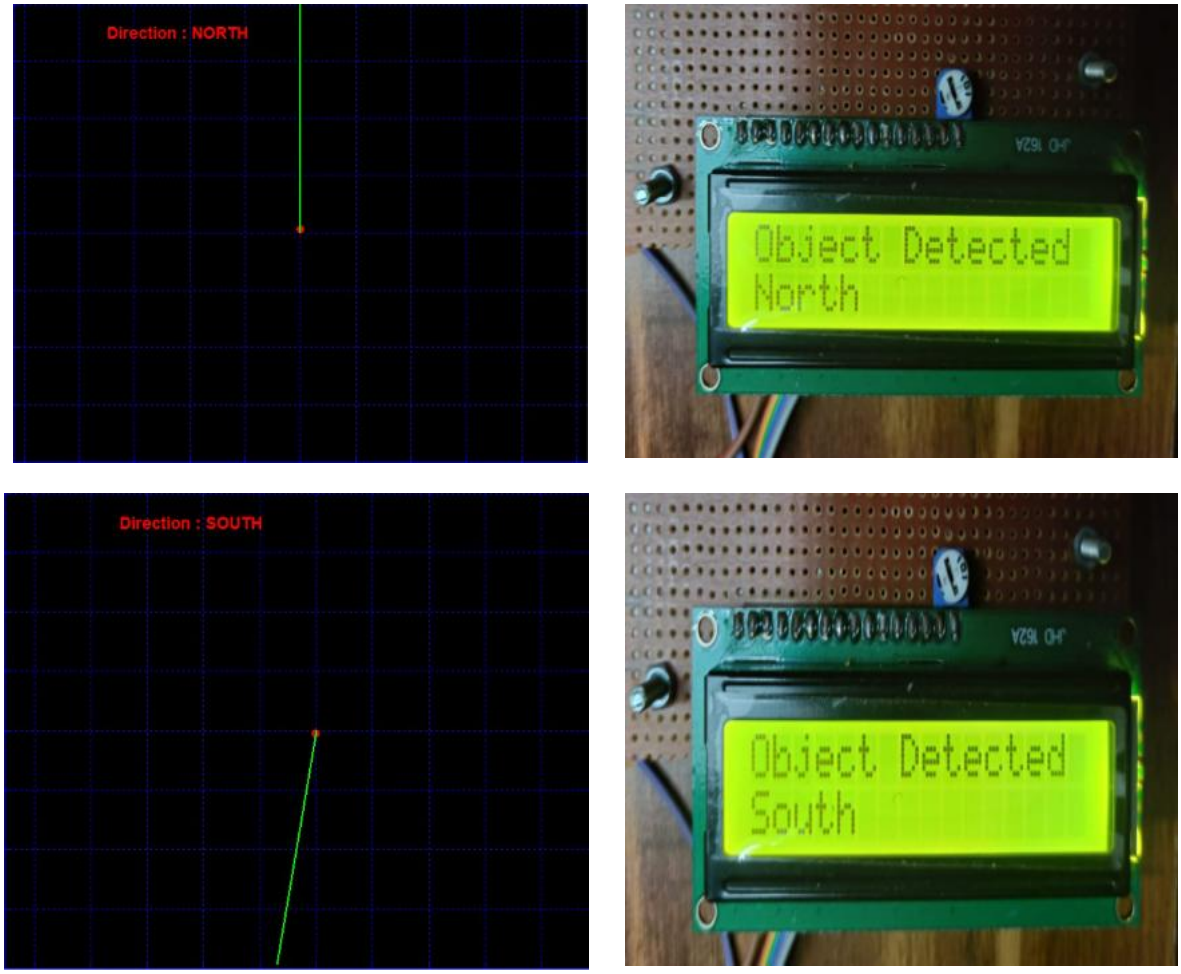

Fig. 6.1. Detection in North and South directions (MATLAB and LCD)

The MATLAB polar plot and LCD display are used to represent the directional result of detection of the **North and South direction**. The objects identified in the polar plot are aligned along the vertical axis with the upwards direction being North (0deg) and the downwards direction being South (180deg). The radial line is used to show how far the object under investigation is located in relation to the sensor. At the same time, the LCD shows the messages "Object Detected North" and Object Detected South as it shows the direction of detection. There is a smooth transition between the North and South detections in continuous scanning. The values that are shown do not fluctuate.

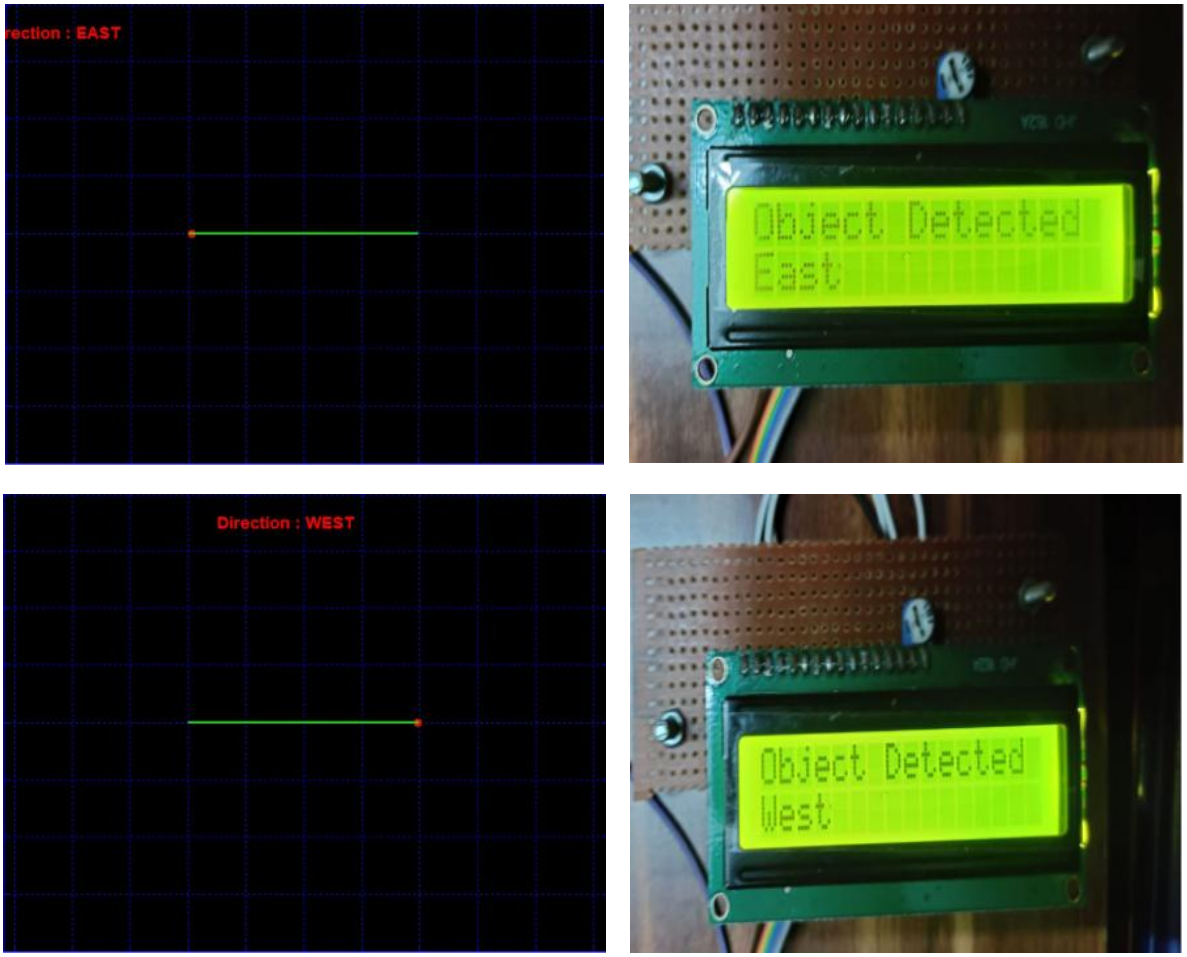

Fig. 6.2. Detection in East and West directions (MATLAB and LCD)

The polarity plot and LCD display are used to indicate the directional detection of the **East and West direction**. The objects found in the polar plot are aligned in the horizontal axis, with the right side of the plot representing East (90deg) and the left side representing West (270deg). The radial line represents the distance of the identified object to the sensor. At the same time, the LCD shows the messages Object Detected East and Object Detected West, which prove the direction of the detection. The graphical display and hardware output is seen to be consistent with one another. The sensing system is well oriented with the detection appearing clearly on the horizontal axis. The system is capable of differentiating between the right and the left directional movements. The change in the East and West detection is seamless when scanning. The output is not unstable and has no abrupt changes or faults.

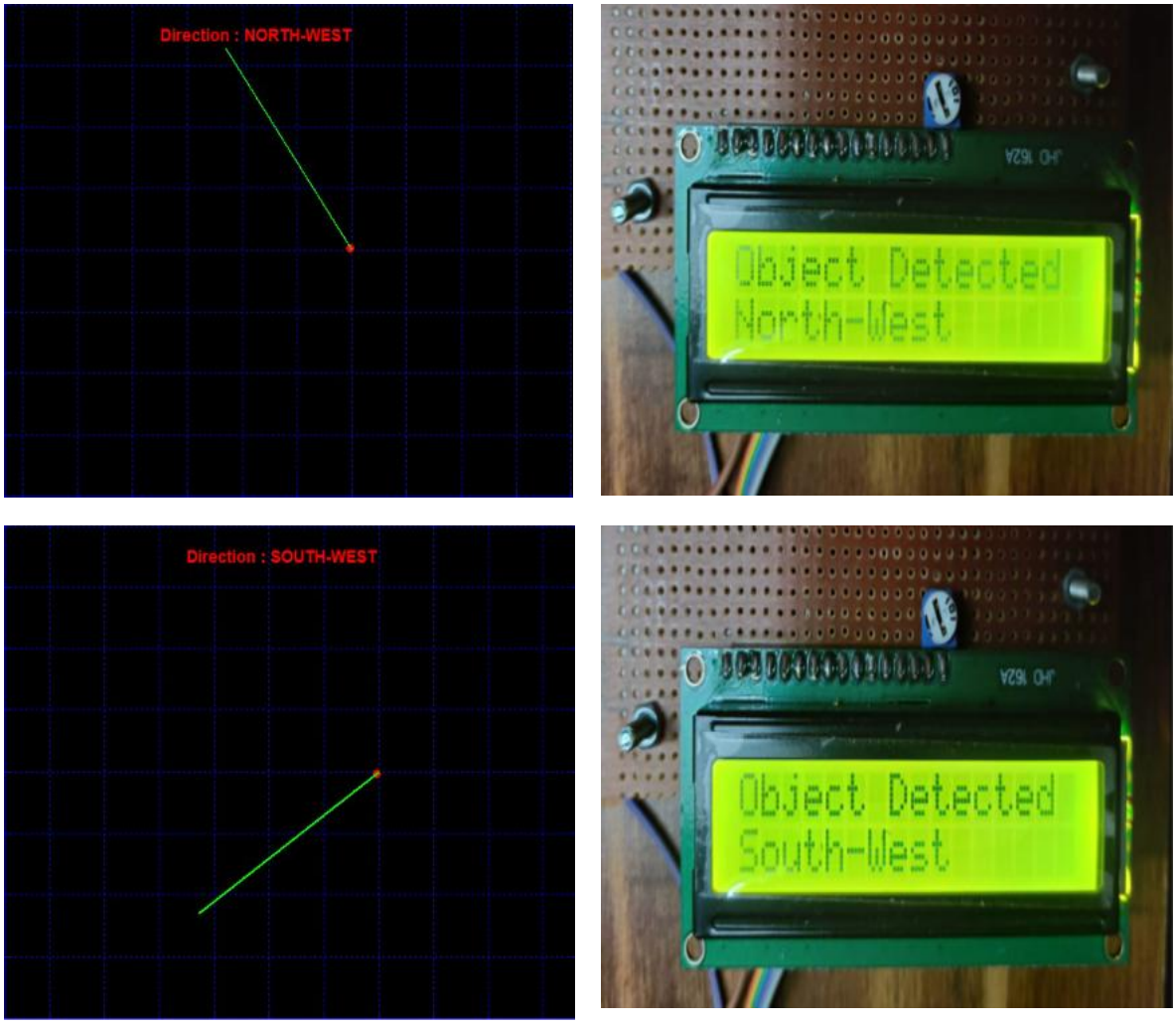

Fig. 6.3. Detection in North-West and South-West directions (MATLAB and LCD)

The findings of directional detection of the **North-West and South-West** directions are plotted with the help of the MATLAB polar plot and LCD display. The objects identified in the polar plot are found in the left diagonal parts, with North-West between 270deg and 360deg and South-West between 180deg and 270deg. The radial line represents the distance of the identified object to the sensor. At the same time, the LCD shows the messages Object Detected North-West and Object Detected South-West, which proves the direction of detection. The graphical display and hardware output is seen to be consistent with one another. The diagonal detection is well differentiated with the primary axes.

The system is accurate in identification of objects in the left-side angular sectors. The border between the North-West and the South-West detection is smooth when scanning continuously. Production is consistent and precise without apparent deviation.

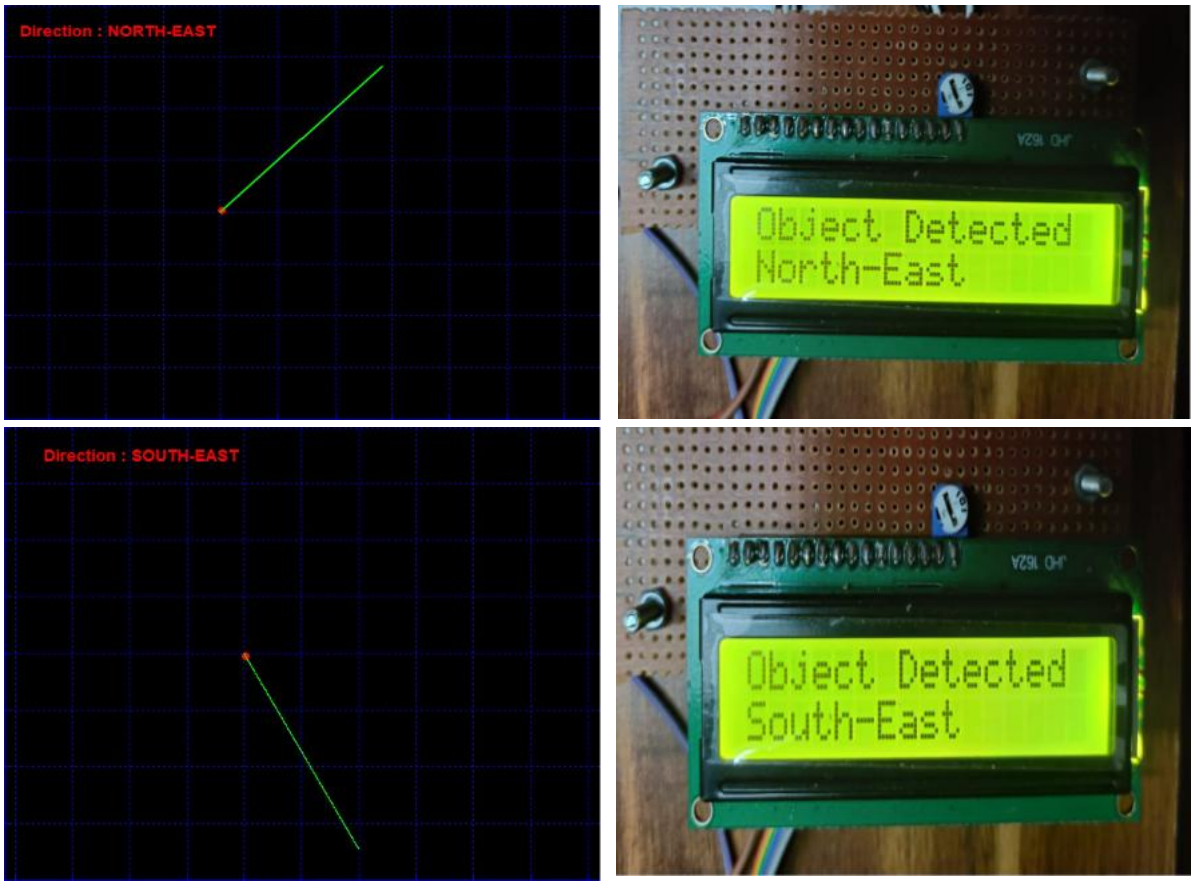

Fig. 6.4. Detection in North-East and South-East directions (MATLAB and LCD)

The MATLAB polar plot and LCD display are used to display the directional detection results of the North-East and South-East direction. The found objects are situated in the right diagonal areas in the polar plot, with North-East situated between 0deg and 90deg and South-East situated between 90deg and 180deg. The radial line represents the distance of the identified object to the sensor. At the same time, the LCD shows the messages that indicate that the object was detected to the North-East and South-East which proves the direction of the detection.

The graphical display and hardware output is seen to be consistent with one another. The primary axes are very differentiated by the diagonal detections. The system is also able to distinguish objects in the right-side angular areas with ease. Continuous scanning has a smooth transition between North-East and South-East detection. The generated output is consistent and precise during the process.

6.4 Cloud IoT Monitoring through Blink Platform

Fig. 6.5. represents the system relayed authenticated intrusion events to Blink IoT cloud dashboard through Wi-Fi. The time stamps and sector were recorded in each detection event. Remote monitoring and access to historical data was facilitated by the cloud interface. It was observed through experimental observations that the latency of cloud update was mainly contingent on network conditions. With average Wi-Fi connection speed, the average transmission delay was within acceptable operational limits in short-range surveillance systems. IoT capability is also integrated, making it very scalable because there is central monitoring of multiple nodes deployed in long border regions.

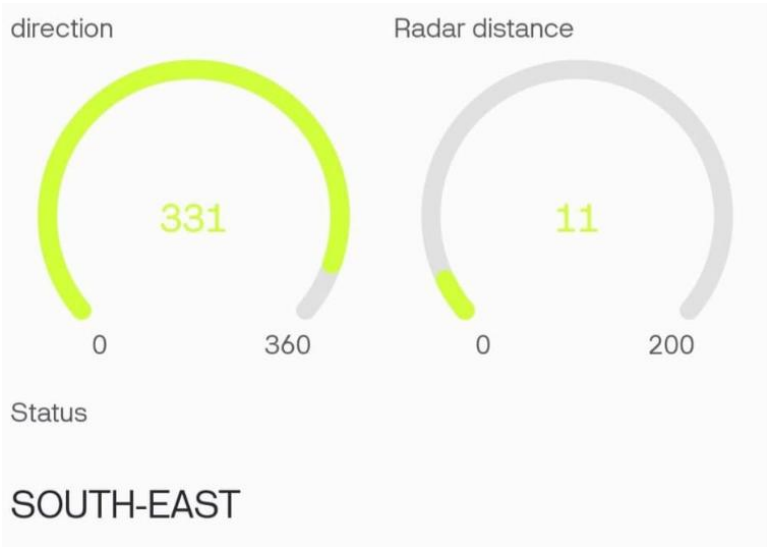

Fig. 6.5. Cloud IoT Blink Platform

6.5 Performance Evaluation of the Proposed System

The general performance of the proposed radar-emulation system was tested on the basis of detection range, latency, consistency, power efficiency, and its ability to reduce false alarms.

Table 6.1 Performance Analysis of Proposed System

Parameter	Observed Value	Remarks
Maximum Reliable Detection Radius	4 m	Stable short-range detection
Average Alert Latency	0.82 s	Includes scanning and processing delay
Detection Consistency	95%	Based on repeated directional trials
Angular Coverage Tested	6 Directions	Indexed sector scanning
Local Display Response Time	~200 ms	Stable output without flicker
False Alarm Reduction	Improved	Due to temporal filtering
IoT Update Capability	Enabled	Real-time cloud logging

These findings show that the system is reliable in the short-range operational factors. The consistency of 95% detection of the test results is a strong repeatability in controlled testing. Table 6.1 summarizes the overall performance metrics of the proposed system.

6.6 Comparative Analysis with Existing Surveillance Systems

Traditional border surveillance systems are human patrol service, fixed CCTV systems, and CNN vision systems that utilize AI. All of these methods have their own drawbacks in the form of limitations in terms of cost, environmental reliance, computational complexity and scalability.

Continuous manpower is needed in human patrol systems and it is prone to fatigue and environmental hazards. The CCTV-based systems are very much dependent on the visibility conditions and can fail in case of fog, dust or in the dark conditions. UAV detection systems are intelligent and require a lot of computational resources, costly cameras, and stable infrastructure.

The ultrasonic system proposed is a cost-effective and modular radar-Guided system that can be used in short-distance detection applications.

Table 6.2 Comparison between Existing and Proposed System

Parameter	Existing Systems (Human Patrol / CCTV / CNN-Based Systems)	Proposed Radar-Assisted System
Cost	High operational or infrastructure cost; manpower-intensive or hardware-expensive	Low-cost embedded implementation using ESP32 and ultrasonic sensor
Environmental Dependency	Highly dependent on lighting, weather, and visibility conditions	Minimally affected by lighting; suitable for low-visibility conditions
Computational Requirement	Moderate to very high (especially CNN-based systems requiring GPUs)	Low computational requirement; suitable for edge processing
Detection Reliability	Variable; depends on human efficiency or image clarity	95% detection consistency in short-range experimental validation
Scalability	Complex expansion due to infrastructure and processing requirements	Easily scalable due to modular and IoT-enabled architecture

The comparative analysis reveals that the proposed system is economically feasible in terms of localized perimeter surveillance especially in resource-limited or temporary deployment conditions. Table 6.2 compares the proposed system with existing surveillance methods based on key parameters.

6.7 Discussion

The experiment confirms the practicality of the ultrasonic radar-emulation method of short-range UAV intrusion detection. Directional awareness was guaranteed by sector-scanning mechanism and false alarms were reduced by temporal filtering due to noise in the environment.

Indexed scanning delay and communication overhead are the main reasons why the average latency is 0.82 seconds. Optimization in the future could involve a faster motor stepping, parallel sensing or a hybrid sensing modules to minimize the detection time. Though the detection radius is restricted to 4 meters because of the limitations of the ultrasonic sensors, the modular design permits the addition of RF, LiDAR or microwave radar modules in the future to enable long range detection. All the results together indicate that the system is cost effective, performance based and deployable and thus is a good prototype system to be used in future research on radar assisted border surveillance.

6.8 Summary

This chapter presented the experimental validation and performance evaluation of the proposed Radar-Assisted Border Surveillance System. Directional detection across six sectors was successfully demonstrated through MATLAB visualization, LCD display output, and Blink IoT cloud logging. Performance analysis confirmed reliable detection within a 4-meter radius, 0.82-second average latency, and 95% detection consistency. Comparative analysis showed clear advantages over conventional surveillance methods in terms of cost, modularity, and environmental robustness. The validated results provide a strong foundation for the concluding chapter, where overall contributions, limitations, and future enhancements of the system will be discussed.

CHAPTER 7
CONCLUSION AND FUTURE WORK

CHAPTER 7
CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project offered design, implementation and validation of a Radar-Assisted Border Surveillance System with UAV Detection that will be used in short-range intrusion detection in remote and resource-limited settings. The system was built on a low-cost ESP32 microcontroller platform coupled with an ultrasonic time-of-flight sensing module and a sector scanning mechanism which is based on a stepper motor. The architecture was able to emulate the radar-like directional detection by indexed angular scanning and real-time signal processing.

It was experimentally verified that the system was reliable in identifying low-altitude intrusions of UAVs within a radius of 4 meters. The attained consistency of detection of 95% during repeat trials indicates consistent behavior in test conditions. The mean alert response of 0.82 seconds is good responsiveness when the systems are to be used in short-range perimeter monitoring. Directional knowledge was well presented with use of three coordinated output systems: MATLAB-based polar display, LCD-based local alert display, and cloud-based event recording using Blink IoT platform.

False alarms generated by the environmental noise and irregular reflection were significantly minimized by the integration of the temporal filtering methods into the embedded firmware. Moreover, the modular design philosophy provided the ease of integration, scalability and low power demands so that the prototype could be deployed in remote border areas where standard surveillance solutions can be either economically or operationally prohibitive.

The comparative analysis between the proposed system and the conventional surveillance methods through human patrol, CCTV monitoring, and CNN-based vision systems indicated that the proposed system is cost-effective, environmental, and has easy infrastructure requirements. Although the detection distance is not much because of the nature of ultrasonic sensing, the prototype provides a solid base in the study of radar-assisted border surveillance in the resource efficient embedded systems. In sum, the project has been able to meet its goals by creating a scalable, functional and low-cost UAV intrusion detection prototype that proves to be practically viable when used in short-range security of the border.

7.2 Future Work

Despite the fact that the proposed system exhibits a high level of reliability in the short-range detection, it is possible to make a number of improvements that will allow broadening its scope of operation and practical use.

The next phase of development could be the implementation of the higher range sensing technologies like the microwave radar modules, RF-based detection systems, or LiDAR sensors to expand the range of detection to more than the current 4-meter maximum. Hybrid sensor fusion with ultrasonic sensing and either vision-based sensing or RF sensing detection modules may further enhance the accuracy of detection and false positives.

Another possible area of improvement is optimization of the scanning mechanism. The use of electronic sector scanning, or multi-sensor setups, would go a long way in lowering the latency and enhancing real-time reactivity. Also, it can be suggested to adopt adaptive scanning algorithms, which are premised on the estimation of threats to improve the efficiency of the system.

Communication wise, secure data transmission guidelines, and encrypted Internet of Things systems may be integrated to enhance the resilience of cybersecurity.

The implementation of the edge-computing-based analytics with the lightweight machine learning models on the ESP32 platform could potentially allow the smart classification of the detected objects, whether it is UAVs or birds or other moving objects.

Solar-powered modules and low-power sleep schedule algorithms are other forms of energy optimization that can be used to enhance the appropriateness to remote border deployments. Moreover, extensive field testing under various environmental conditions such as fog, rain and wind disturbances would give more information on system robustness.

To sum up, although the current prototype confirms that radar-emulation-based UAV intrusion detection can be used to perform short-range surveillance, future research and technological improvements can turn the system into a more sophisticated, long-range, and intelligent method of border security.

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CHAPTER 8

PUBLICATIONS

PUBLICATIONS

Nanjundeswari M, Naveen G, Nisita M, Vinothkumar U, (2026) 'Sector-Scanning Ultrasonic Radar-Emulation System for Short-Range UAV Intrusion Detection', 6th International Conference on Trends in Material Science and Inventive Materials - (ICTMIM 2026).

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Acceptance Letter

Paper ID	Paper Title	Author's Name
ICTMIM-237	Sector-Scanning Ultrasonic Radar-Emulation System for Short-Range UAV Intrusion Detection	Vinothkumar U, Nanjundeswari M, Naveen G, Nisita M

Dear Authors,

On behalf of the Committee, we take great pleasure in inviting you to attend the "6th International Conference on Trends in Material Science and Inventive Materials (ICTMIM 2026)" which is going to be held during 8-10, April 2026 at Rohini College of Engineering and Technology, Kanyakumari, Tamil Nadu

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